

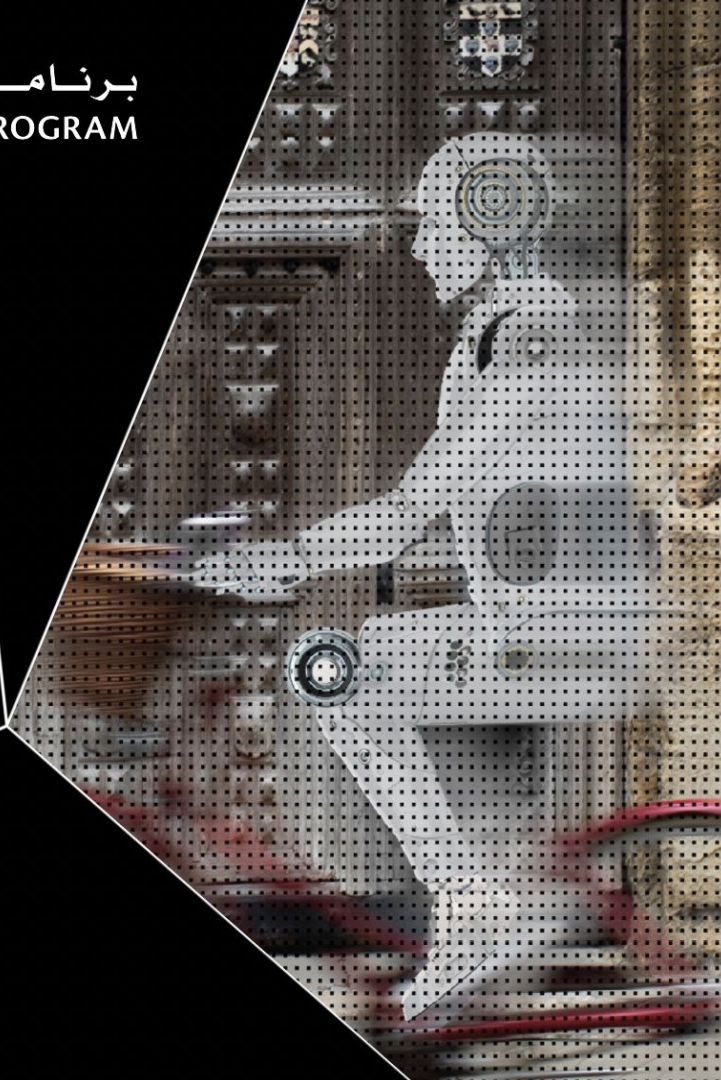


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THE ARTIFICIAL INTELLIGENCE PROGRAM

Identifying Opportunities for AI

Farah, Dhiyaa and Firas



Objectives for the day

Part 1 (11:00 am - 12:30 pm)

- AI or Not Case Studies Workshop
- Samples of Capstone Projects

Part 2 (1:30 - 3 pm)

- Brainstorming & Problem Definition
- Ideation & Innovation

AI or Not Case Studies

- Group Case Study Discussion (30 minutes)
 1. *What specific problem is being addressed?*
 2. *Is this problem best suited for an AI-driven solution?*
 3. *What type of data is available, and is it sufficient, representative, and good enough to train a robust AI model?*
 4. *Could a simpler or more traditional solution achieve comparable outcomes with lower risk?*
- Presentations and Q&A (30 minutes)



AI or Not Case Study Example

numbers.txt

John Doe: +971503955103
Jane Doe: +971567273678
Robert Smith: +971508883221
Maria Garcia: +971507835841
...



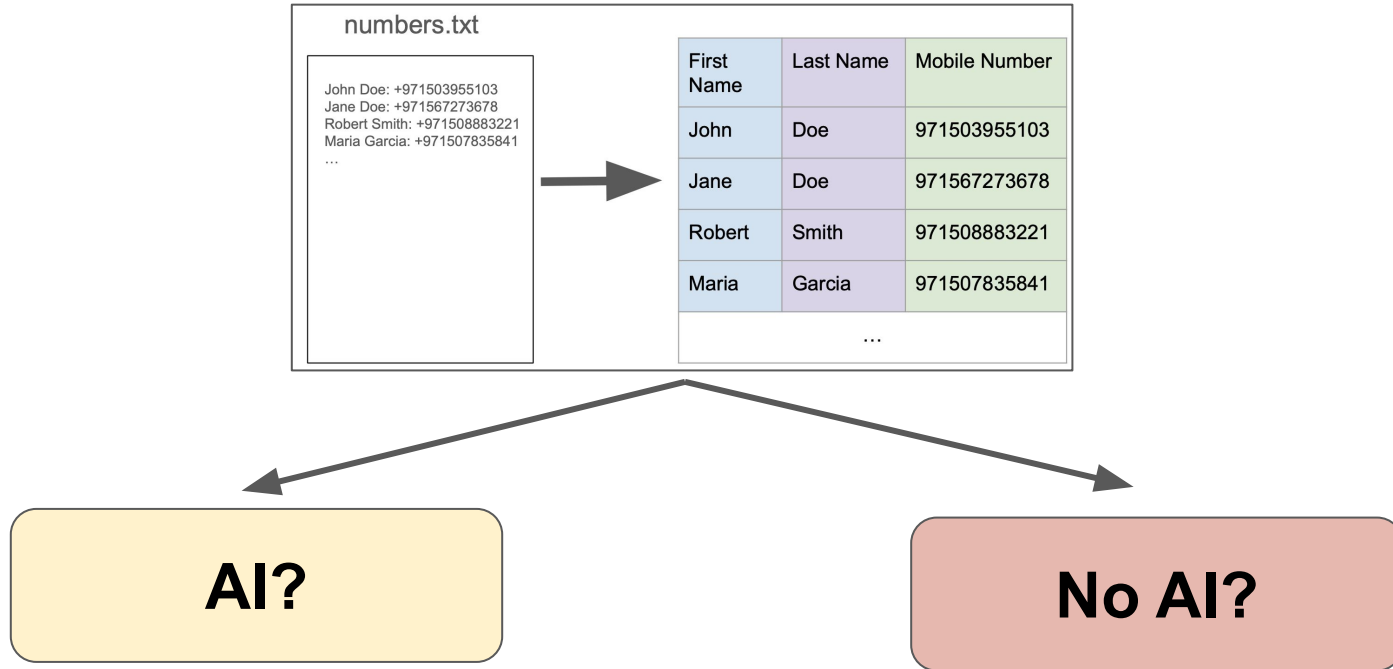
First Name	Last Name	Mobile Number
John	Doe	971503955103
Jane	Doe	971567273678
Robert	Smith	971508883221
Maria	Garcia	971507835841
...		

1. What specific problem is being addressed?

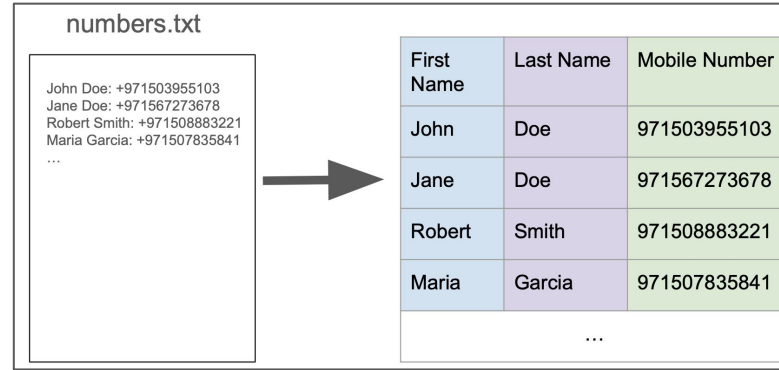
The problem is automatically extracting names and phone numbers from a **structured text file**, where the information appears in a fixed format and context.



AI or Not Case Study Example



AI or Not Case Study Example



2. *Is this problem best suited for an AI-driven solution?*

No. This task involves identifying patterns that are well-defined (e.g., names and phone number formats).

While AI could handle more complex extraction or possibly organizing unstructured data, this specific problem is better suited for a traditional pattern-matching approaches using predefined patterns.

No AI?



AI or Not Case Study Example

3. What type of data is available, and is it sufficient, representative, and good enough to train a robust AI model?

If you were to train an AI model:

- You'd need a **large, labeled dataset** with examples of the mapping from line to first name/last name/mobile number.
- For this task, the available data is **likely** to be sufficient for robust AI training assuming the inputs are strictly organized in the same way, **even though it is not recommended to use AI.**

numbers.txt

John Doe: +971503955103
Jane Doe: +971567273678
Robert Smith: +971508883221
Maria Garcia: +971507835841
...

First Name	Last Name	Mobile Number
John	Doe	971503955103
Jane	Doe	971567273678
Robert	Smith	971508883221
Maria	Garcia	971507835841
...		

numbers.txt

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numbers.txt

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Maria Garcia: +971507835841
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➔

First Name	Last Name	Mobile Number
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Maria	Garcia	971507835841
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numbers.txt

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First Name	Last Name	Mobile Number
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Maria	Garcia	971507835841
...		

AI or Not Case Study Example

4. *Could a simpler or more traditional solution achieve comparable outcomes with lower risk?*

Yes. A well-crafted regular expression (pre-defined text patterns that are matched) can effectively and efficiently extract phone numbers in most common formats. It's:

- Faster to implement
- Easier to maintain
- Less resource-intensive
- Transparent and explainable

```
import re
import pandas as pd

text = '''
John Doe: +971503955103
Jane Doe: +971567273678
Robert Smith: +971508883221
Maria Garcia: +971507835841
'''

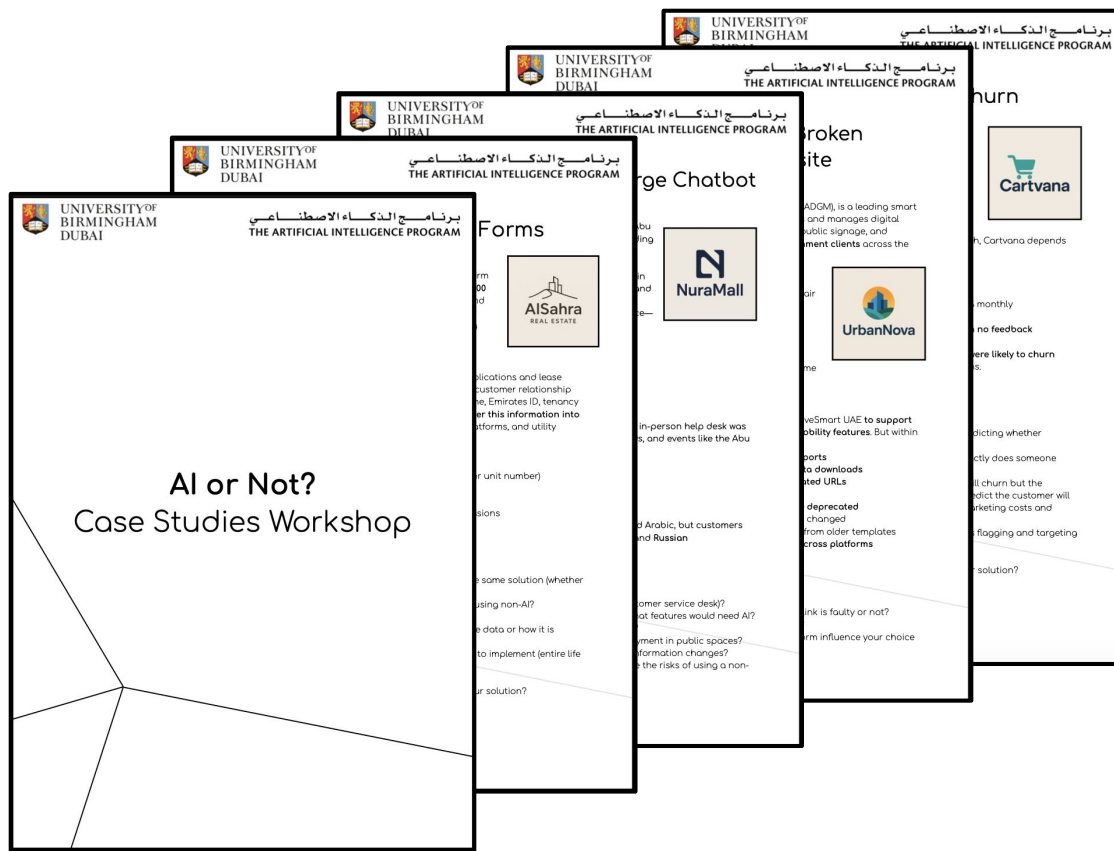
# Pattern to extract First Name, Last Name, Phone Number
pattern = r'(\w+)\s+(\w+):\s+(\+\d{12})'

# Use regex to find all matches in the text
matches = re.findall(pattern, text)

# Create Structured Table
df = pd.DataFrame(matches, columns=["First Name", "Last Name", "Phone Number"])
```

	First Name	Last Name	Phone Number
0	John	Doe	+971503955103
1	Jane	Doe	+971567273678
2	Robert	Smith	+971508883221
3	Maria	Garcia	+971507835841





AI or Not Slide Template

Make a copy of this template and fill it out with your team's responses.



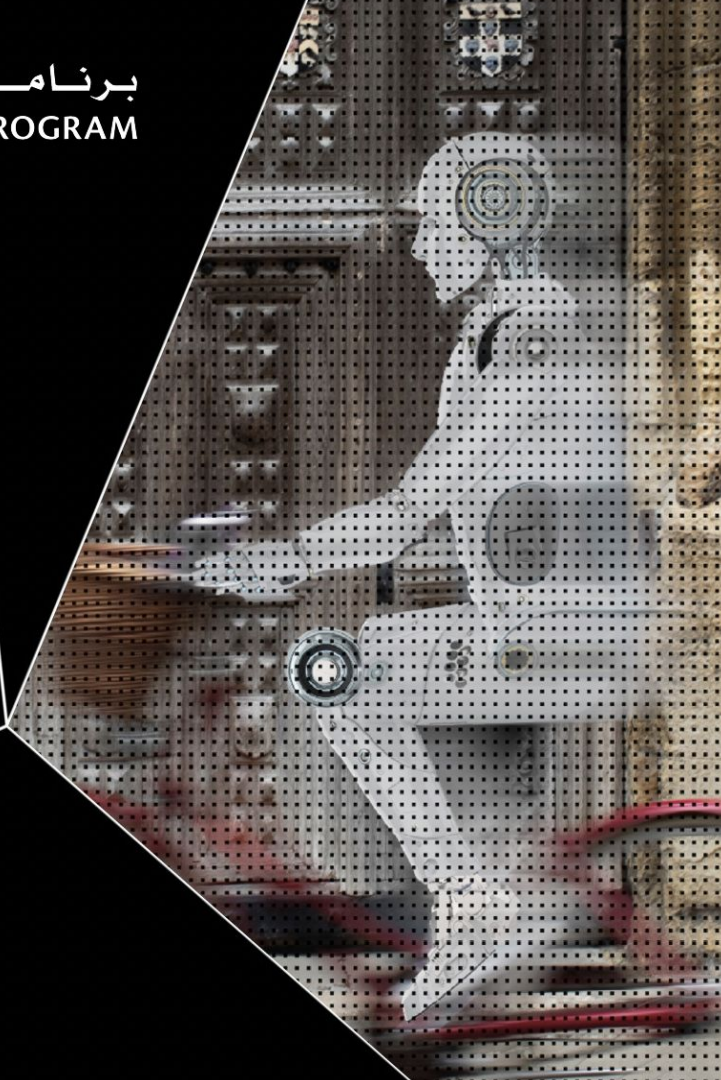


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Lunch time



Samples of Capstone Projects

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Supply Chain Optimization Using AI Techniques



Group Members (Team 15):

Balqis AlBreiki
Sultan AlAmeri
Amna AlHammadi
Mohamed Al Ali

Date: September 17, 2024

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Image Generated using AI

AI-Driven Crime Detection: Leveraging BERT to Combat Illegal Activities on Social Media for Enhanced Public Safety in the UAE
Sep 2024

Team Members	
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Brainstorming & Ideation

Round 1 (30 minutes)

Speed-brainstorming of potential problems and write them on post-its per table

*Looking for DMCC, e& and construction sites/
operations, ENERGY SECTOR*



Brainstorming & Ideation

Round 2 (30 minutes)

Choose a set of three problems and suggest and flesh out AI solutions

What dataset would you need?

Is the idea feasible?

Is the project impactful?

Discuss with mentors

1-minute pitch per idea per team and receive feedback from audience



Brainstorming & Ideation



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Brainstorming Self-Evaluation

Idea 1 Title

Task Definition (Define the following clearly and sufficiently)

What exactly is the problem?

End Users

What is the impact of the project?

What are some pre-existing non-AI approaches to solving the problem?

Why is AI suitable/necessary for solving this task?

What kind of data do you need in terms of type, quantity, robustness, and is it available/collectible?

What risks or concerns are there associated with the project? (Logistical, time, privacy, regulation, etc.)



Brainstorming & Ideation

Round 3 (30 minutes)

Select your top two final ideas

Submit the ideas on this google form with your team information:

<https://forms.gle/oHShsg44JAT1zhNn6>



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